Listing 10.16 Title and unit attributes of a legend

```
var title = "Legend";
var numberFormat = d3.format(".4n");
var units = "Units";

//other code

legend.title = function(newTitle) {
  if (!arguments.length) return title;
  title = newTitle;
  return this;
};

legend.unitLabel = function(newUnits) {
  if (!arguments.length) return units;
  units = newUnits;
  return this;
};

legend.formatter = function(newFormatter) {
  if (!arguments.length) return numberFormat;
  numberFormat = newFormatter;
  return this;
};
```

These are added right after var scale inside the d3.svg.legend function.

All these functions are added right after legend.scale.

We'll use these new properties in our updated legend drawing code shown in listing 10.17. This new code draws SVG `<line>` elements at each breakpoint, and foregoes the rotated text in favor of more readable, shortened text labels at each breakpoint. It also adds two new `<text>` elements, one above the legend that corresponds to the value of the `title` variable and one at the far right of the legend that corresponds to the `units` variable.

Listing 10.17 Updated legend drawing code

```
gSelection.selectAll("line")
     .data(data)
     .enter()
     .append("line")
     .attr("x1", function (d) {return xScale(d.domain[0]);})
     .attr("x2", function (d) {return xScale(d.domain[0]);})
     .attr("y1", 0)
     .attr("y2", size[1] + 5)
     .style("stroke", "black")
     .style("stroke-width", "2px");

gSelection.selectAll("text")
     .data(data)
     .enter()
     .append("g")
     .attr("transform", function (d) {
            return "translate(" + (xScale(d.domain[0])) +","
                          + (size[1] + 20) + ")";
     })
     .append("text")
```

This follows your existing code to draw the legend <rect> elements, and updates the text.

Each line is drawn at the breakpoint and drawn a little lower to "point" at the breakpoint value.